TOLCID 4% INJECTION

Description and Composition
TOLCID 4% INJECTION is a colourless to slightly yellow clear solution. Each ml contains 40mg Tolfenamic acid as its active ingredient and 10mg Benzyl alcohol as preservative.

Pharmacodynamics
Tolfenamic acid (N-(2-methyl-3-chlorophenyl)anthranilic acid) is a nonsteroidal anti-inflammatory drug from the fenamate group. Tolfenamic acid has anti-inflammatory, analgesic and antipyretic properties. The anti-inflammatory activity of tolfenamic acid is due to the inhibition of cyclooxygenase leading to a reduction in the synthesis of prostaglandins and thromboxane, which are important inflammatory mediators.

Pharmacokinetics
In dogs, tolfenamic acid is rapidly absorbed by injection. After injection of 4 mg tolfenamic acid / kg, the mean peak plasma concentration of approximately 4 µg/mL (subcutaneous) and 3 µg/mL (intramuscular) is reached 2 hours after administration. In cats, absorption is very rapid. After administration by injection of 4 mg of tolfenamic acid / kg, the average maximum plasma concentration of 3.9 µg / mL is reached in one hour. More than 99% of tolfenamic acid is bound to plasma proteins. In dogs, only tolfenamic acid and its conjugates with glucuronic acid are found in the urine. The hydroxylated metabolites and their conjugates are mainly excreted by the kidneys. Tolfenamic acid in unchanged form and its glucuronides are mainly excreted in the bile. In addition, tolfenamic acid undergoes intensive enterohepatic cycling. Tolfenamic acid is distributed in all organs with high concentrations in plasma, digestive tract, liver, lungs and kidneys. However, the concentration in the brain is low. In dogs with renal insufficiency, the elimination of tolfenamic acid is not modified and there is no accumulation.

In cattle and pigs, tolfenamic acid injected intramuscularly at a dose of 2 mg/kg is rapidly absorbed from the injection site, giving, after one hour, mean peak plasma concentrations of approximately 1. 4 µg/mL in cattle and 2.3 µg/mL in pigs. The volume of distribution is approximately 1.3 L/kg. Tolfenamic acid is highly bound to plasma albumin (>97%). Tolfenamic acid is distributed in all organs with high concentrations in plasma, digestive tract, liver, lungs and kidneys. However, the concentration in the brain is low. Tolfenamic acid and its metabolites hardly cross the placental barrier. Distribution of tolfenamic acid involves extracellular fluids where concentrations achieved are similar to those in plasma, in both healthy and inflamed peripheral tissues. It also appears in milk in an active form, mainly in curds. Tolfenamic acid undergoes extensive enterohepatic cycling leading to prolonged plasma concentrations. The elimination half-life varies from 3 to 5 hours in pigs and from 8 to 15 hours in cattle. In cattle and pigs, tolfenamic acid is eliminated mainly unchanged in faeces (~30%) and urine (~70%).

Indication
Cattle: Adjuvant treatment of pneumonia by improving general conditions and runny nose. Adjuvant treatment of acute mastitis.
Pigs: Adjuvant treatment of Metritis Mastitis Agalactia syndrome.
Dogs: Treatment of inflammation associated with musculoskeletal disorders and reduction of post-operative pain.
Cats: Adjuvant treatment of upper respiratory diseases, in combination with antimicrobial treatment, if necessary.

Recommended Dose
Cats and dogs: 4mg Tolfenamic acid per kg body weight (i.e. 1ml per 10kg body weight), in a single injection, repeat if necessary, once after 24 to 48 hours depending on clinical assessment.

Or a single injection of 1ml per 10kg body weight, followed by oral treatment with tablets.
Administration in dogs by intramuscular or subcutaneous route. Administration in cats only by subcutaneous route. To reduce post-operative pain, it is advisable to perform a pre-operative injection, as a premedication, one hour before induction of anesthesia.

Cattle:
Treatment of inflammation associated with respiratory diseases: 2mg Tolfenamic acid per kg body weight (i.e.1ml per 20kg body weight) intramuscularly in the neck region. The treatment can be repeated once after 48 hours.
Treatment of inflammation associated with mastitis: 4mg Tolfenamic acid per kg body weight (i.e. 1ml per 10kg body weight), in a single intravenous injection. In case of intravenous administration, the product should be injected slowly. At the first signs of intolerancethe injection should be stopped.

Pigs: 2mg Tolfenamic acid per kg body weight (i.e. 1ml per 20kg body weight), in a single injection by intramuscular route.

Do not exceed the dose of 20 mL per injection site.
For low weight animals, it is advisable to use an insulin syringe in order to administer a precise dose.

Route of Administration
Cattle: Intramuscular or intravenous routes.
Pigs: Intramuscular route.
Dogs: Intramuscular or subcutaneous routes.
Cats: Subcutaneous route.

Contraindication
Do not use in case of cardiac pathology.
Do not use in case of impaired liver function or acute renal failure.
Do not use in case of digestive ulceration or bleeding, nor in case of blood dyscrasia.
Do not inject intramuscularly in cats.
Do not use in case of known hypersensitivity to the active substance or to any of the excipients.
Do not use in dehydrated, hypovolemic or hypotensive animals (due to possible risk of increased renal toxicity)
Do not administer other anti-inflammatory drugs, steroidal or non-steroidal, simultaneously or within 24 hours of treatment.

Warning and Precautions
Special warning for target species:
NSAIDs may cause inhibition of phagocytosis and therefore treatment of inflammation associated with bacterial infections should be combined with appropriate antibacterial therapy.

Special precautions for use:
<u>Special precautions for use in animals</u>
Use of the drug in animals less than 6 weeks of age, or in older animals, may involve additional risks. If such use cannot be avoided, these animals may require a reduced dosage and careful clinical monitoring is essential.
Consideration should be given to the decreased metabolism and excretion in these animals.
Avoid simultaneous administration of potentially nephrotoxic drugs.
It is best not to administer the product to cats undergoing general anesthesia until they have fully recovered.
Do not exceed the recommended dose and duration.
The pain reduction scale after pre-operative administration may be influenced by the severity and duration of the operation.
Although animals requiring anti-inflammatory therapy and exhibiting chronic renal failure can be treated without the need for dosage adjustment, this treatment is contraindicated in cases of acute renal failure.
In the event of adverse effects (anorexia, vomiting, diarrhoea, presence of blood in the faeces) occurring during treatment, seek advice from your veterinarian and the possibility of interrupting treatment should be studied.
Respect the rules of asepsis when administering the product.

<u>Special precautions to be taken by the person administering the veterinary medicinal product to animals</u>
The product may cause skin sensitization.
People with known hypersensitivity to nonsteroidal anti-inflammatory drugs (NSAIDs) or to any of the excipients should avoid contact with the drug.
Administer the drug with caution to avoid accidental self-injection.
In case of accidental self-injection, seek medical advice immediately and show the package insert or the label to the physician.
The product may cause skin or eye irritation.
Avoid all contact with skin and eyes.
In case of accidental contact, wash the exposed area with clean water.
Seek medical attention if irritation persists.
Wash hands after usee.

Interaction with Other Medicinal Product
Do not administer at the same time as other non-steroidal anti-inflammatory drugs (NSAIDs) or respect an interval of 24 hours between the 2 administrations.
Other NSAIDs, diuretics, anticoagulants and substances with high affinity for plasma proteins could compete for binding sites and cause toxic effects.
Do not administer at the same time as anticoagulants.
Avoid simultaneous administration of potentially nephrotoxic drugs.
Do not administer simultaneously with glucocorticoids.

Pregnancy and Lactation
Dogs and cats: Do not use during pregnancy.
Cattle and pigs:
Pregnancy: use should only be made after assessment of the benefit/risk ratio established by the responsible veterinarian.
Lactation: the product can be used during lactation.

Side Effects
Syncope after rapid intravenous injection may occur in cattle infrequently.
A temporary increase in thirst and/or diuresis may occur in very rare cases.
In most cases, these signs cease spontaneously after treatment.
Diarrhea and vomiting may very rarely occur during treatment.
If these signs persist, treatment should bediscontinued.
Local reactions at the injection site may occur in very rare cases.
The frequency of side effects is defined as follows:
- very common (adverse effects in more than 1 in 10 animals treated)
- frequent (between 1 and 10 animals out of 100 treated animals)
- uncommon (between 1 and 10 animals in 1,000 animals treated)
- rare (between 1 and 10 animals out of 10,000 animals treated).
- very rare (less than one animal in 10,000 animals treated, including isolated cases)..

Symptoms and Treatment of Overdose
At high doses, neurological disorders have been observed. In case of overdose, symptomatic treatment should be administered.

Storage Condition
Store below 30ºC. Protect from light.

Shelf life
3 years. Following withdrawal of the first dose, use the product within 24 hours. Any unused material should be discarded.

Withdrawal Periods
Cattle:
Intramuscular injection:
Meat and offal: 12 days.
Milk: zero hours.
Intravenous injection:
Meat and offal: 4 days.
Milk: 24 hours.
Pigs: Meat and offal: 16 days.

Packing
20ml, 50ml, 100ml, 250ml.

Revision date: 21.9.2022

	Manufacturer & Product Registration Holder:
Range Pharma	Range Pharma Sdn Bhd No.1, Jalan TSB 11, Taman Industri Sg. Buloh, 47000 Sg. Buloh, Selangor Darul Ehsan, Malaysia Tel: 603-61568708 Fax: 603-61568707 E-mail: info@rangepharma.com